

# Cyclotomic Polynomials

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## 1 Why do we care about Cyclotomic Polynomials?

1. Every finite Division Ring is in fact a field.
2. Weak Dirichlet Theorem on Primes in Arithmetic Progressions: for any integer  $n$ , there are infinitely many primes of the form  $kn + 1$ .
3. Zsigmondy's Theorem: for any integers  $a > b > 0$  and  $n > 1$ , there exists a prime that divides  $a^n - b^n$  but does not divide  $a^k - b^k$  for any  $k < n$ , with some exceptions.

## 2 What is a Cyclotomic Polynomial?

1. Definition: The  $n$ -th cyclotomic polynomial  $\Phi_n(x)$  is defined as the monic polynomial whose roots

are the primitive  $n$ -th roots of unity. Explicitly,

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k,n)=1}} (x - e^{2\pi i k/n}).$$

### Example 2.1

$\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$  for a prime  $p$ .  $\Phi_1(x) = x - 1$ .  $\Phi_2(x) = x + 1$ .  $\Phi_3(x) = x^2 + x + 1$ .  $\Phi_4(x) = x^2 + 1$ .

2.  $x^n - 1$  can be factored as

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Why?

3. Its degree is  $\phi(n)$ .

4.  $\Phi_n(x)$  is a monic polynomial with integer coefficients.

*Proof.* Use induction on  $n$ .  $\Phi_n(x) \mid x^n - 1$  in  $\mathbb{C}[x]$ , its quotient is  $f(x) = \prod_{d|n, d < n} \Phi_d(x)$ , which is in  $\mathbb{Z}[x]$  by inductive hypothesis. Consider the long division of  $x^n - 1$  by  $f(x)$  in  $\mathbb{Z}[x]$ .  $x^n - 1 = f(x)q(x) + r(x)$ . So  $f(x)(q(x) - \Phi_n(x)) = -r(x)$ . So  $r(x) = 0$  and  $q(x) = \Phi_n(x) \in \mathbb{Z}[x]$ .  $\square$

5.  $\Phi_n(x)$  is irreducible in  $\mathbb{Q}[x]$ .

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*Proof.* Let  $f(x)$  be the irreducible proper factor of  $\Phi_n(x)$ .

**Case 1:** There exists  $m$  and  $p \nmid n$  such that  $f(\zeta_n^m) = 0$  but  $f(\zeta_n^{pm}) \neq 0$ .

Let  $\Phi_n(x) = f(x)g(x)$ . So  $\zeta_n^m$  is a root of  $g(x^p)$ . Since  $f$  is irreducible, it must be the minimal polynomial of  $\zeta_n^m$ . So  $f(x) \mid g(x^p)$  in  $\mathbb{Z}[x]$ . Thus,  $f(x) \mid g(x^p) = g(x)^p$  in  $\mathbb{F}_p[x]$ . So there exists an irreducible  $l(x) \mid f(x) \mid g(x)^p$  so  $l(x) \mid g(x)$ . So  $l(x)^2 \mid \Phi_n(x) \mid x^n - 1$ . Since  $p \nmid n$ ,  $x^n - 1$  is separable but  $l(x)^2$  is not. Contradiction.

**Case 2:** For any root  $\zeta_n^m$  of  $f(x)$  and any  $p \nmid n$ ,  $\zeta_n^{mp}$  is also a root of  $f(x)$ . In this case, we claim that  $f(x) = \Phi_n(x)$ . Indeed: if for any  $t$  that is coprime to  $n$ , there exists  $k$  coprime to  $n$  such that  $t \equiv mk \pmod{n}$ . (both  $t$  and  $m$  have inverse) Therefore,  $\zeta_n^t = \zeta_n^{mk} = \zeta_n^{mp_1 p_2 \cdots p_i} \cdot p_i$  where  $p_1$  to  $p_i$  are all coprime to  $n$ . Thus,  $\zeta_n^t$  is a root of  $f(x)$ .

□

6. Thus, we see that  $\mathbb{Q}(\zeta_n)$  is the splitting field of  $\Phi_n(x)$ . Since  $\Phi_n(x)$  is irreducible and Galois,  $\mathbb{Q}(\zeta_n)$  must be Galois. And its Galois group is  $(\mathbb{Z}/n\mathbb{Z})^\times$

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### 3 get to know it

1. Find  $\Phi_n(x)$  for  $N \leq 8$ .
2. Find  $\Phi_{2^m}(x)$
3. If  $n$  is odd, prove  $\Phi_{2n}(x) = \Phi_n(-x)$
4. In general,

$$\Phi_{pn}(x) = \begin{cases} \Phi_n(x^p) & p \mid n \\ \frac{\Phi_n(x^p)}{\Phi_n(x)} & p \nmid n \end{cases}$$

5. Prove: integer  $a > 1$ , if  $\Phi_n(a) \mid a - 1$ , then  $n = 1$ .  
In fact,  $|x - 1|^{\phi(n)} < \Phi_n(x) < |x + 1|^{\phi(n)}$  for all  $x$ .
6. Prove:  $\Phi_n(x) \mid \frac{x^n - 1}{x^d - 1}$  in  $\mathbb{Z}[x]$  for all proper divisors  $d$  of  $n$ .

### 4 Wedderburn's Little Theorem

**Theorem:** Every finite division ring is a field.

1. If  $D$  is a finite division ring, then its center is a finite field.  $Z = \mathbb{F}_q$  for some  $q$ .
2.  $D$  is a vector space over  $\mathbb{F}_q$  with dimension  $n$ . So  $|D| = q^n$ .
3. Any centralizer is also a vector space, thus  $|C_D(a)| = q^d$  for some  $d \mid n$ .

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4. Remove the 0 and look at them as a group. By class formula:

$$q^n - 1 = |Z| + \sum \frac{|G|}{|Stab_G(r)|} = q - 1 + \sum \frac{q^n - 1}{q^d - 1}.$$

5. So  $\Phi_n(x) \mid (q - 1)$ . So  $n = 1$  and  $D = Z$  is a field.

## 5 $\Phi_n(x)$ and order of an element mod $p$

1. Let  $a$  be an integer. If  $p \mid \Phi_n(x)$ , then  $p \mid a^n - 1$
2. Moreover, if  $p \nmid n$ ,  $p \nmid a^d - 1$  for all  $d < n$ . Why? (separability comes into play here)
3. Thus, if  $p \nmid n$ , then  $p \mid \Phi_n(a)$  iff the order of  $a$  in mod  $p$  is  $n$ .
4. Furthermore,  $n \mid p - 1$  Why? Order of  $a$  divides  $p - 1$ !
5. Thus, there are infinitely many primes  $\equiv 1 \pmod{n}$ .

## 6 Zsigmondy's Theorem

1. Our goal is to find a prime such that  $p \mid a^n - 1$  but not  $a^d - 1$  for any  $d < n$ . So  $p$  must divide  $\Phi_n(a)$ .

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2. If we have a prime  $p \mid \Phi_n(a)$ , then we wish  $p \nmid n$  to ensure that  $p \nmid a^d - 1$  for all  $d < n$ . (separability consideration)
  3. So given  $a$ , is there a prime such that  $p \mid \Phi_n(a)$  but  $p \nmid n$ ?
  4. Mostly, yes.
  5. Assume not. Let  $p_1, p_2, \dots, p_k$  be all the prime divisors of  $n$ . Then  $\Phi_n(a)$  is consisted of  $p_1, p_2, \dots, p_k$ .
  6. By lifting the exponent lemma, we have  $v_{p_i}(\Phi_n(a)) \leq v_{p_i}\left(\frac{a^n - 1}{a^{n/p_i^e} - 1}\right) + v_{p_i}(n) \leq v_{p_i}(n)$ . So  $(a - 1)^{\phi(n)} < \Phi_n(a) \leq n$ . This is clearly wrong when  $a$  is large.